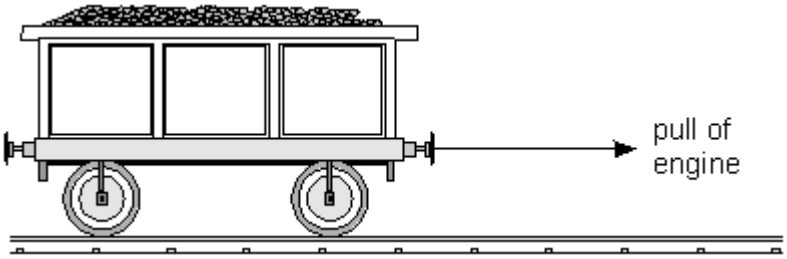
Q1.



(a) A railway engine is being used to try to pull a wagon along a level track. The wagon's brakes are on, and the wagon does not move.

(i) Draw **one** arrow on the diagram to show the direction of the force which prevents the wagon from moving.

1 mark

(ii) Is the force which prevents the wagon from moving **greater than**, **equal to** or **less than** the pull of the engine?

.....

1 mark

(b) (i) When the wagon's brakes are off, the engine pulls the wagon forwards. A frictional force also acts on the wagon. In what direction does the frictional force act?

1 mark

(ii) The pull of the engine is 5000 N. When the wagon's speed is increasing, how large is the frictional force?
Tick the correct box.

zero ☐

between 0 and 5000 N ☐

5000 N ☐

more than 5000 N ☐

1 mark

(c) After a while, the wagon travels at a steady speed. The engine is still pulling with a force of 5000 N.

How large is the frictional force now?
Tick the correct box.

zero ☐

between 0 and 5000 N ☐

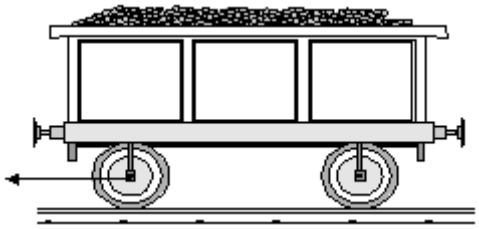
5000 N ☐

more than 5000 N ☐

1 mark
Maximum 5 marks

Q1.

(a) (i)



the mark is for an arrow pointing to the left as shown
the arrow may be anywhere on the diagram

accept an arrow pointing to the left, drawn in the space beneath the question

1 (L5)

(ii) equal to

accept 'equal'

1 (L5)

(b) (i) backwards

accept 'in the opposite direction to the movement' ***or*** *'in the opposite direction'* ***or*** *'to the left'*

accept an arrow drawn pointing to the left either on the diagram, if labelled clearly, or in the space beneath the question

1 (L5)

(ii) between 0 and 5000 N ✓

if more than one box is ticked, award no mark

1 (L5)

(c) 5000 N ✓

if more than one box is ticked award no mark

1 (L5)

[5]